# PRODUCT USER MANUAL

## Ocean Physical and Biogeochemical reanalysis

## NWSHELF_MULTIYEAR_PHY_004_009

## NWSHELF_MULTIYEAR_BGC_004_011

**Issue:** 1.4

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**Approval Date: November 2025**

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | Aug 2020 | All | Merged PUM_004_009 and PUM_004_011.<br>Updated for Dec 2020 release. | M. Tonani and I. Ascione | C. Derval |
| 1.1 | Jan 2021 | All | Updated for May 2021 Release | M. Tonani and I. Ascione | C. Derval |
| 1.2 | Aug 2021 | II.3 and II.4 | Details on vertical grid and vertical interpolation. | M. Tonani and I. Ascione | C. Derval |
| 1.3 | Feb 2022 | II.2 | removed reference to MDT | I. Ascione A. Saulter | C. Derval |
| 1.4 | May 2025 | | Update for November 2025 release (merge of MY and MYint) and general adjustment (about datasets and data access) | R.Renshaw, T.Brüning | |

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## GLOSSARY AND ABBREVIATIONS

| | |
|---|---|
| CF | Climate Forecast (convention for NetCDF) |
| CMEMS | Copernicus Marine Environment Monitoring Service |
| ECMWF | European Centre for Medium Range Weather forecast |
| FTP | Protocol to download files |
| IOC | Intergovernmental Oceanographic Commision of UNESCO |
| MDS | Marine Data Store |
| Meridional Velocity | West to East component of the horizontal velocity vector |
| MFC | Monitoring and Forecasting Centre |
| MY | Multi Year |
| NetCDF | Network Common Data Form |
| NWS | North-West Shelf |
| OPENDAP | Open-Source Project for a Network Data Access Protocol. Protocol to download subset of data from a n-dimensional gridded dataset (ie: 4 dimensions: lon-lat, depth, time) |
| PC | Production Centre |
| PU | Production Unit |
| PUM | Product User Manual |
| QUID | Quality Information Document |
| RMS | Root mean square |
| SSC | Sea surface currents |
| SSH | Sea surface height |
| SSS | Sea surface salinity. |
| SST | Sea surface temperature |

| | |
|---|---|
| Zonal Velocity | South to North component of the horizontal velocity vector |

## DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

# I INTRODUCTION

## I.1 Summary

This document is the user manual for the North West-European Shelf reanalysis products:

**NWSHELF_MULTIYEAR_PHY_004_009;**

**NWSHELF_MULTIYEAR_BGC_004_011.**

These products are generated by a coupled physical-biogeochemistry model system.

The first release of the time series covers the period 01/01/1993-31/12/2019. The time series is extended twice yearly for the full reanalysis. Additionally there are monthly (interim) updates, so that data is available to within 3 months of the present day (see also product improvements pages https://marine.copernicus.eu/services-portfolio/product-improvements/).

The products are published on and available through the MDS after automatic and human quality control. Downloaded files downloaded are in NetCDF format and follow CF-1.7 convention.

The reanalysis system is described in the Quality Information Documents (QUIDs):

https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-009.pdf;
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-011.pdf.

More detailed information can be obtained from the Copernicus Marine Service Team.

## I.2 History of changes

| Date | Description of changes and impacted product |
|---|---|
| December 2020 | New times series from 01/01/1993 to 31/12/2019.<br><br>Data assimilation of Phytoplankton Functional Types (PFT) |
| May 2021 | Additional interim datasets |
| November 2025 | Merge of MY and MY interim datasets, temporal (interim) extension of additional datasets |

# II PRODUCT DESCRIPTION:

## II.1 General Information about products

| | |
|---|---|
| **Product name** | NWSHELF_MULTIYEAR_PHY_004_009<br>NWSHELF_MULTIYEAR_BGC_004_011 |
| **Geographical coverage** | 20°W → 13°E<br>40°N → 65°N<br>The outermost model grid points at the Atlantic and Baltic boundaries are masked.<br>Baltic boundary: all the grid points East of 10°E are masked.<br>Atlantic boundaries: grid points north of 64°18'48" N; west of -18°19'12" E and south of 40°43'58" N are masked (outermost 10-grid points). |
| **Variables PHY_004_009** | Potential temperature<br>Salinity<br>Sea surface height<br>Horizontal velocity (eastward and northward components)<br>Bottom temperature<br>Mixed layer depth |
| **Variables BGC_004_011** | Volume Beam Attenuation Coefficient of Radiative Flux<br>Concentration of Chlorophyll a<br>Mole Concentration of Dissolved Oxygen<br>Mole Concentration of Nitrate<br>Mole Concentration of Phosphate<br>Mole Concentration of Phytoplankton Expressed as Carbon<br>Net Primary Production<br>Surface partial pressure of carbon dioxide<br>pH reported on total scale<br>mass concentration of diatoms expressed as chlorophyll in sea water<br>mass concentration of dinoflagellates expressed as chlorophyll in sea water<br>mass concentration of nanophytoplankton expressed as chlorophyll in sea water |

| | |
|---|---|
| | mass concentration of picophytoplankton expressed as chlorophyll in sea water |
| **Update frequency** | The product is updated twice a year (June and December) providing six-month extensions of the full reanalysis time series to within 6 months of the present day. Monthly interim extensions are provided by the penultimate Tuesday of each month to bring the reanalysis time series to within 3 months of present day. |
| **Available time series** | The first release is from 01/01/1993 to 31/12/2019. The product is updated twice a year providing six-month extension (see product improvements pages http://marine.copernicus.eu/services-portfolio/product-improvements/). The interim part of the time series extends from the end of the full processed reanalysis to three months before the present. |
| **Temporal resolution** | Monthly, Daily means and hourly instantaneous |
| **Horizontal resolution** | ~7km<br>1/9°longitude x 1/15°latitude |
| **Number of vertical levels** | 24 |
| **Delivery mechanism** | Copernicus Marine Toolbox |
| **Format** | NetCDF4 (FTP) – CF1.7 |

### II.2 Details of the datasets

| **NWSHELF_MULTIYEAR_PHY_004_009** | |
|---|---|
| Daily(**dm**) and monthly (**mm**) means fields of:<br>• Potential temperature, salinity, zonal and meridional velocity (3D);<br>• Sea Surface Height (SSH), mixed layer depth, and sea bottom temperature (2D)<br>2D Hourly (hi) fields of:<br>• Potential temperature, salinity, zonal and meridional velocity, sea surface height, and sea bottom temperature | |
| **DATASETS** | Variables name in the NetCDF file and Unit<br>Long_name<br>CF Standard_name |
| cmems_mod_nws_phy-uv_my_7km-3D_P1D-m | **vo** [$ms^{-1}$]<br>Northward Current Velocity in the Water Column<br>northward_sea_water_velocity |

| | |
|---|---|
| cmems_mod_nws_phy-uv_my_7km-3D_P1M-m<br>cmems_mod_nws_phy-uv_my_7km-2D_PT1H-i | **uo** [$ms^{-1}$]<br>Eastward Current Velocity in the Water Column<br>eastward_sea_water_velocity |
| cmems_mod_nws_phy-t_my_7km-3D_P1D-m<br>cmems_mod_nws_phy-t_my_7km-3D_P1M-m<br>cmems_mod_nws_phy-sst_my_7km-2D_PT1H-i | **thetao** [degrees_C]<br>Sea Water Potential Temperature<br>sea_water_potential_temperature |
| cmems_mod_nws_phy-s_my_7km-3D_P1D-m<br>cmems_mod_nws_phy-s_my_7km-3D_P1M-m<br>cmems_mod_nws_phy-sss_my_7km-2D_PT1H-i | **so** [1e-3]<br>Sea Water Salinity<br>sea_water_salinity |
| cmems_mod_nws_phy-bottomt_my_7km-2D_P1D-m<br>cmems_mod_nws_phy-bottomt_my_7km-2D_P1M-m<br>cmems_mod_nws_phy-bottomt_my_7km-2D_PT1H-i | **bottomT** [degrees_C]<br>Sea floor potential temperature<br>sea_water_potential_temperature_at_sea_floor |
| cmems_mod_nws_phy-mld_my_7km-2D_P1D-m<br>cmems_mod_nws_phy-mld_my_7km-2D_P1M-m<br>cmems_mod_nws_phy-mld_my_7km-2D_PT1H-i | **mlotst**[m]<br>Ocean mixed layer thickness defined by density<br>ocean_mixed_layer_thickness_defined_by sigma_theta |
| cmems_mod_nws_phy-ssh_my_7km-2D_P1D-m<br>cmems_mod_nws_phy-ssh_my_7km-2D_P1M-m<br>cmems_mod_nws_phy-ssh_my_7km-2D_PT1H-i | **zos**[m]<br>Sea surface height above geoid<br>sea_surface_height_above_geoid |

| NWSHELF_MULTIYEAR_BGC_004_011 | |
|---|---|
| Daily(**dm**) and monthly (**mm**) mean fields of:<br>• 3D biogeochemical variables<br>• 2D surface partial pressure of CO2 | |
| **DATASETS** | Variable's name in the NetCDF file and Unit<br>Long_name<br>CF Standard_name |
| cmems_mod_nws_bgc-kd_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-kd_my_7km-3D_P1M-m | **attn** [m-1]<br>Volume Beam Attenuation Coefficient of Radiative Flux in Sea Water<br>volume_beam_attenuation_coefficient_of_radiative_flux_in_sea_water |
| cmems_mod_nws_bgc-chl_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-chl_my_7km-3D_P1M-m | **chl** [mg m-3]<br>Mass Concentration of chlorophyll in Sea Water<br>mass_concentration_of_chlorophyll_a_in_sea_water |
| cmems_mod_nws_bgc-o2_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-o2_my_7km-3D_P1M-m | **o2** [mmol m-3]<br>Mole Concentration of Dissolved Oxygen in Sea Water<br>mole_concentration_of_dissolved_molecular_oxygen_in_sea_water |
| cmems_mod_nws_bgc-no3_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-no3_my_7km-3D_P1M-m | **no3** [mmol m-3]<br>Mole Concentration of Nitrate in Sea Water<br>mole_concentration_of_nitrate_in_sea_water |
| cmems_mod_nws_bgc-po4_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-po4_my_7km-3D_P1M-m | **po4** [mmol m-3]<br>Mole Concentration of Phosphate in Sea Water<br>mole_concentration_of_phosphate_in_sea_water |
| cmems_mod_nws_bgc-phyc_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-phyc_my_7km-3D_P1M-m | **phyc** [mmol m-3]<br>Mole Concentration of Phytoplankton Expressed as Carbon in Sea Water<br>Mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water |
| cmems_mod_nws_bgc-pp_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-pp_my_7km-3D_P1M-m | **nppv** [mg m-3 day-1]<br>Net Primary Production<br>net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water |

| | |
|---|---|
| cmems_mod_nws_bgc-spco2_my_7km-2D_P1D-m<br>cmems_mod_nws_bgc-spco2_my_7km-2D_P1M-m | **spco2** [Pa]<br>Surface partial pressure of carbon dioxide in sea water<br>surface_partial_pressure_of_carbon_dioxide_in_sea_water |
| cmems_mod_nws_bgc-ph_my_7km-3D_P1D-m<br>cmems_mod_nws_bgc-ph_my_7km-3D_P1M-m | **ph**[1]<br>Sea Water pH reported on total scale<br>Sea_water_ph_reported_on_total_scale |
| cmems_mod_nws_bgc-pft_my_7km-3D-diato_P1M-m<br>cmems_mod_nws_bgc-pft_my_7km-3D-diato_P1D-m | **diato [mg m-3]**<br>mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water<br>mass concentration of diatoms expressed as chlorophyll in sea water |
| cmems_mod_nws_bgc-pft_my_7km-3D-dino_P1M-m<br>cmems_mod_nws_bgc-pft_my_7km-3D-dino_P1D-m | **dino [mg m-3]**<br>mass_concentration_of_dinoflagellates_expressed_as_chlorophyll_in_sea_water<br>mass concentration of dinoflagellates expressed as chlorophyll in sea water |
| cmems_mod_nws_bgc-pft_my_7km-3D-nano_P1M-m<br>cmems_mod_nws_bgc-pft_my_7km-3D-nano_P1D-m | **nano [mg m-3]**<br>mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water<br>mass concentration of nanophytoplankton expressed as chlorophyll in sea water |
| cmems_mod_nws_bgc-pft_my_7km-3D-pico_P1M-m<br>cmems_mod_nws_bgc-pft_my_7km-3D-pico_P1D-m | **pico [mg m-3]**<br>mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water<br>mass concentration of picophytoplankton expressed as chlorophyll in sea water |

The model **bathymetry** and **Land-Sea** mask information are stored in the static datasets:

| ***DATASETS*** | Variable's name in the NetCDF file and Unit<br>Long_name<br>*CF Standard_name* |
|---|---|
| cmems_mod_nws_phy_my_7km-3D_static<br>cmems_mod_nws_bgc_my_7km-3D_static | **mask**<br>Land-sea mask<br>sea_binary_mask |
| | **depth** [m]<br>depth |

| | |
|---|---|
| | depth |
| | **deptho** [m]<br>Bathymetry<br>sea_floor_depth_below_geoid |
| | **deptho_lev_interp**<br>Deepest wet grid point of the interpolated 3D fields<br>model_level_number_at_sea_floor |

## II.3 Details on some parameters

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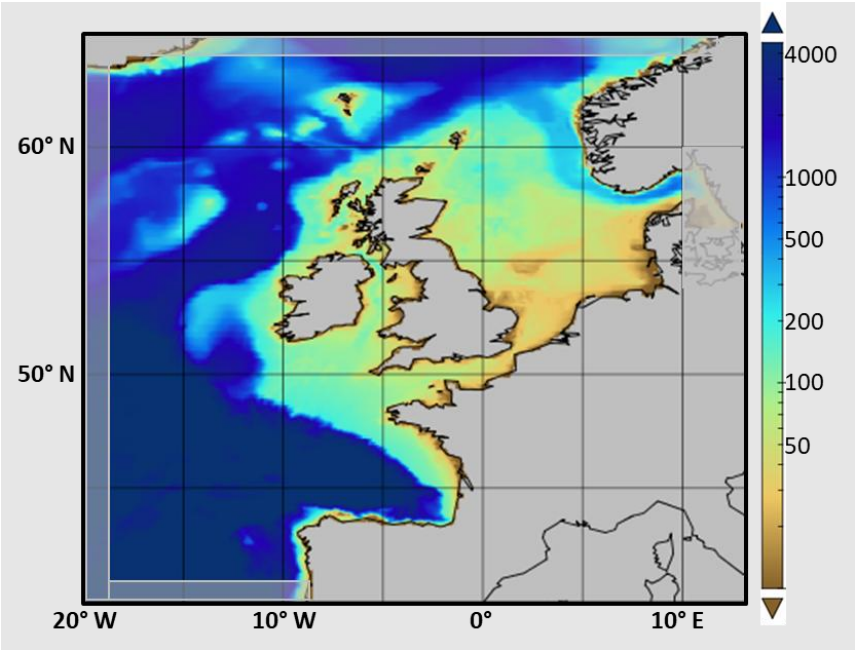
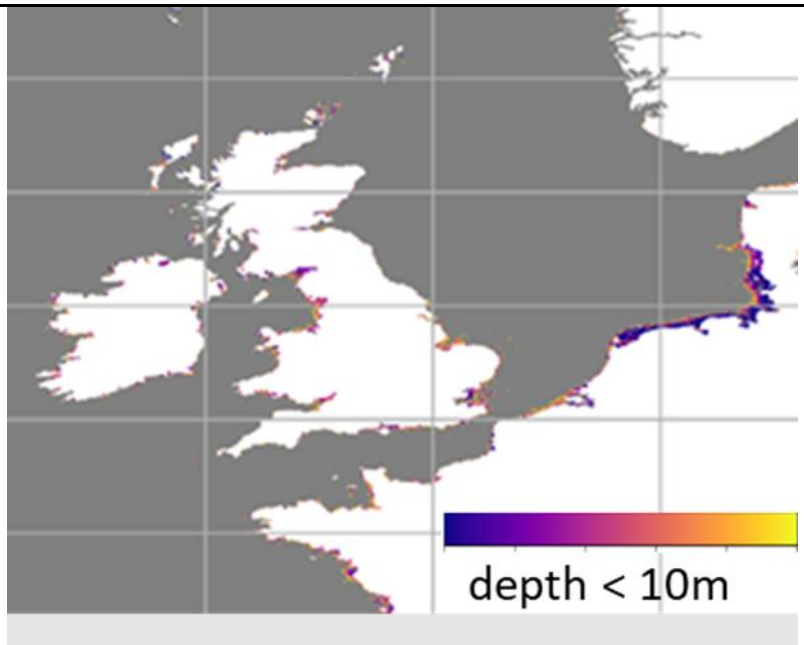
|---|---|
| **mlotst** [m] | ocean_mixed_layer_thickness_defined_by_sigma_theta. It is the depth where the density increase compared to density at 3m depth corresponds to a temperature decrease of 0.2°C in local surface conditions. |
| **zos** (m) | sea_surface_height_above_geoid: The geoid is a surface of constant geopotential with which mean sea level would coincide if the ocean were at rest. The SSH is the difference between the actual sea surface height at any given time and place, and that which it would have if the ocean were at rest. |
| **bottomT** (degrees_C) | sea_water_potential_temperature_at_sea_floor is output directly from the native model; it is not interpolated. It means that bottom and the last wet point of the 3D temperature dataset could be different. |
| **deptho_lev_interp** | Deepest wet grid point of the Interpolated 3D fields. |

**II.4 Product System Description**

| | |
|---|---|
| Geographical domain/ Resolution | NORTH WEST EUROPEAN SHELF (20°W - 13°E; 40°N - 65°N)<br>1/15° longitude x 1/9° latitude (297x 375 grid points)<br>  <br><br>*NWS bathymetry and the domain of the PHY_004_009 and BGC_004_011 products. The grey translucent areas show where the model output is masked.*<br><br>The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.<br><br>The model is a nested regional ocean model configuration, coupled with the biogeochemical model. PHY and BGC products are delivered on an identical horizontal grid.<br><br>A **minimum depth of 10m** is imposed in the ocean model for considering the sea level tidal excursion. Products below the surface in areas where the bathymetry is less than 10m could be affected by this modelling choice. |




1) What is done:

We take every bathymetric grid point < 10 m in the domain and reassign it with a depth of 10 m, coloured point in the figure above. These grid points are interpolated back to their original depth before delivering the file to the CMEMS catalogue. Areas such as the Wadden Sea are particularly affected (see below reference for a case study in the Wadden sea).

2) Why it is done:

This is because version 3.6 of NEMO does not have wetting and drying capability. If the bathymetry were not modified, it would mean that with the large tides that exist in this domain the model would not be able to handle the case when the tide goes out.

The minimum depth prevents such a situation from occurring and keeps the model as a whole stable.

3) What is the likely effect.

The prorogation of tides and surges are inherently related to the water depth. The speed is proportional to $\sqrt{gH}$ where $H$ is the water depth and thus as the model water depth is deeper than reality the prorogation will be incorrect.

In cases where the seabed is exposed then this process is completely absent from the model by design (there will be water in the model where in reality there is none). It is thus not recommended to the user to directly use the model fields in regions where there are extensive areas of bathymetry less than 10 m in depth. If a model is to be nested within this product, then it is recommended that the boundaries of the nested domain are sufficiently far from such areas to reduce their impact.

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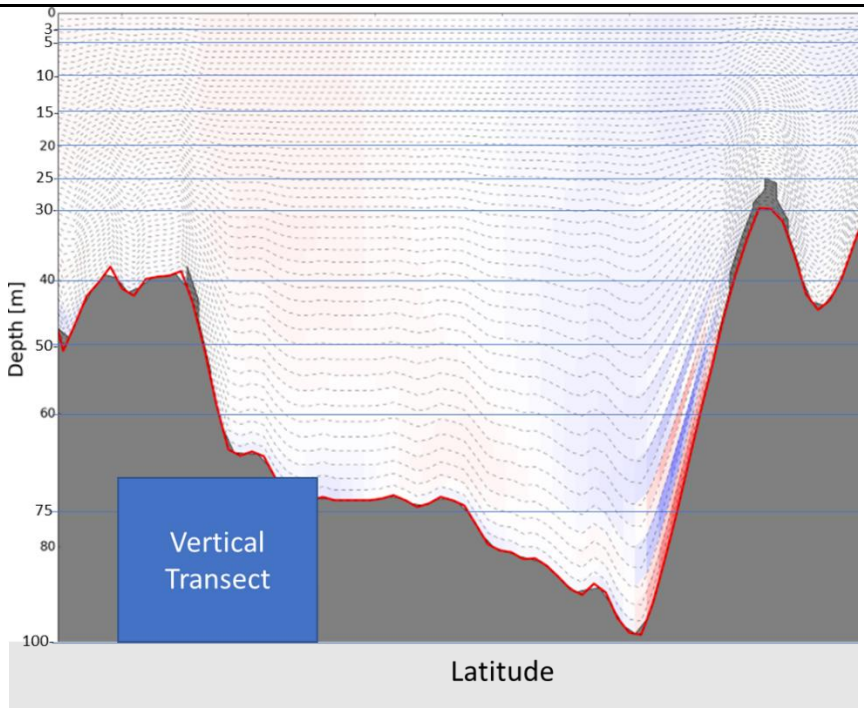
| | 4) Wetting and Drying is implemented in NEMO 4.<br>This specific problem is addressed with the wetting and drying module in NEMO 4 which will be part of future development of this operational Copernicus product.<br>The interested reader is directed to the recent paper on its development: https://doi.org/10.1016/j.ocemod.2020.101708<br>Enda O'Dea, Michael J. Bell, Andrew Coward, Jason Holt, Implementation and assessment of a flux limiter based wetting and drying scheme in NEMO, Ocean Modelling, Volume 155,2020. |
| Horizontal grid | ~7km (1/9°longitude x 1/15°latitude) |
| Vertical grid | The products are interpolated for users' convenience, into 24 standard IOC geopotential levels: [0 (surface), 3, 10, 15, 20, 30, 50, 75, 100, 125, 150, 200, 250,300, 400, 500, 600, 750, 1000, 1500, 2000, 3000, 4000, 5000 m ].<br><br>The surface level is not interpolated, it is the first model level. It has a thickness of 1m where the bathymetry is deeper than 50m. Its thickness is less than 1m where the bathymetry is shallower than 50m.<br><br>The model native vertical grid has Hybrid S-σ-z-coordinates. The terrain-following coordinates system uses a stretching function to maintain near uniform vertical resolution at the surface. The number of S-σ-z levels is 51 almost everywhere in model domain. It means that the thickness of the model vertical grid cells is smaller in shallow area compare to the deep part of the basin.<br>The thickness of the model native vertical grid varies from 0.3m (in areas with a bathymetry < 50m) to 99m (where the bathymetry is > 4000m). |

| | |
|---|---|
| |  <br>*Vertical transect: it shows the differences between the model native vertical grid (dotted line) and the CMEMS products interpolated at standard geopotential depths (blue lines). The model bathymetry (red lines) doesn't match exactly the bathymetry (shaded grey area) due to numerical constrains.*<br>The bottom temperature (bottomT) is output directly from the native model, it is not interpolated.<br>All other CMEMS NWS-MFC products are vertically interpolated from the 51 native terrain following coordinates to fixed depths. |
| Model Version | ERSEM15.06 (biogeochemical model) coupled to NEMO 3.6 (physical model) via FABM |
| Tides | Tidal forcing is included both on the open boundary conditions via a Flather radiation boundary conditions (Flather, 1976) and through the inclusion of equilibrium tide from a tidal model of the North-East Atlantic (Flather, 1981).<br>15 tidal constituents: M2, S2, N2, K2, K1, O1, P1, Q1, M4, MS4, L2, T2, S1, 2N2, MU2, NU2. |
| Atmospheric forcing | ECMWF – ERA5 |
| Assimilation scheme | 3D Var FGAT. NEMOVAR. |
| Assimilated observations | • RAN & INT: ESA-CCI SST L2/L3 products (ATSR and AVHRR)<br>• RAN: in-situ SST observations from ICOADS (via HadIOD)<br>• RAN: temperature and salinity subsurface profiles from EN4 (HadIOD) |

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| | • INT: in situ SST and sub-surface temperature and salinity profiles from Met Office MetDB database (observations received in near real-time over the GTS) and CMEMS INS TAC<br>• RAN & INT: satellite Chlorophyll PFTs from ACRI-ST: OCEANCOLOUR_ATL_BGC_L3_NRT_009_011 |
| Boundaries conditions | T, S, SSH, barotropic u and v at the Atlantic lateral boundary: UK Met Office Global reanalysis v13 |
| | T, S, barotropic u and v at the Baltic lateral boundary: BALTICSEA_REANALYSIS_PHY_003_011 to end of 2022. For later times (RAN & INT) we use a climatology which is daily means calculated from this same dataset for 2020-2022. |
| | Nitrate, Phosphate and Oxygen at the Atlantic and Baltic boundaries: WOA 2013v2 monthly climatology<br>Alkalinity and dissolved inorganic carbon from GLODAP v2 climatology<br>Nitrogen deposition at surface from EMEP project (time-varying). |
| | Daily timeseries of river discharge, nutrient loads (nitrate, phosphate, silicate, ammonia), alkalinty (total alkalinity, bioalkalinity, dissolved organic carbon), and oxygen from 1991-2017. Timeseries data are produced from an updated version of the river dataset used in Lenhart et al. (2010) combined with climatology of daily discharge data from the Global River Discharge Data Base (Vörösmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by Young and Holt (2007). Interannual changes in the series are small and so the 2017 time series is repeated for subsequent years. |
| Initial conditions | The model run started on 01/01/1991 from initial conditions from the previous version of the NWS reanalysis. 01/01/1991-31/12/1992 is not a product as considered model spin up time. |
| Coupling | One-way on-line coupling: physics forcing the biogeochemistry.<br>Coupling frequency: at every model time step. |
| Bathymetry | GEBCO 1 arc-minute data together with a variety of other local data sources, supplied by the North West-European Shelf Operational Oceanographic System (NOOS) partners. |

## II.5 Processing information

### *II.5.1 Update Time*

Full Reanalysis: twice a year in June (up to December of the preceding year) and December (up to June of that year), providing time series extension of six months.

Reanalysis-interim (RAN-INT): monthly (on the next-to-last Tuesday of each month).

### *II.5.2 Time coverage*

The first release is from 01/01/1993 to 31/12/2019. The product is updated twice a year, providing six-month extension of the time series (see product improvements pages http://marine.copernicus.eu/services-portfolio/product-improvements/). Extensions of the full reanalysis system are provided every June (up to December of the preceding year) and December (up to June of that year), so that the full reanalysis is never more than 1 year behind real time.

The RAN-INT timeseries starts from the last available day of the reanalysis described above and is updated monthly (on the next-to-last Tuesday of each month) to be no more than three months behind real-time. For instance, the update to end of February is provided in May.

### *II.5.3 Time averaging*

Daily mean value: the fields are daily means over a day, midnight to midnight, centered at noon. The mean is computed over 25 hours (from 00:00 of T to 00:00 UTC of T+1) to remove (de-tide) the tidal signal.

Monthly mean values: the fields are monthly means over the calendar month (first to last day of the month, calculated from hourly means).

### *II.5.4 Time variable in netCDF files*

Time is expressed in seconds since epoch 1970-01-01, as follows:

```
double time(time) ;
time:units = "seconds since 1970-01-01T00:00:00Z" ;
time:long_name = "Validity time" ;
time:standard_name = "time" ;
time:calendar = "Gregorian" ;
time:axis = "T" ;
```

# III FILES NOMENCLATURE AND FORMAT

## III.1 File Format

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

## III.2 Files Nomenclature

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

## III.3 File size

| NWSHELF_MULTIYEAR_PHY_004_009 | | |
|---|---|---|
| **DATASET NAME** | **NAME OF FILE** | **DIMENSION [MB]** |
| cmems_mod_nws_phy-bottomt_my_7km-2D_P1D-m | metoffice_foam1_amm7_NWS_BED_dm{{yyyymmdd}}.nc | 0.1 |
| cmems_mod_nws_phy-uv_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_CUR_dm{{yyyymmdd}}.nc | 2.6 |
| cmems_mod_nws_phy-mld_my_7km-2D_P1D-m | metoffice_foam1_amm7_NWS_MLD_dm{{yyyymmdd}}.nc | 0.1 |
| cmems_mod_nws_phy-s_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_SAL_dm{{yyyymmdd}}.nc | 1.3 |

| | | |
|---|---|---|
| cmems_mod_nws_phy-ssh_my_7km-2D_P1D-m | metoffice_foam1_amm7_NWS_SSH_dm{{yyyymmdd}}.nc | 0.1 |
| cmems_mod_nws_phy-t_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_TEM_dm{{yyyymmdd}}.nc | 1.6 |
| cmems_mod_nws_phy-bottomt_my_7km-2D_P1M-m | metoffice_foam1_amm7_NWS_BED_mm{{yyyymm}}.nc | 0.2 |
| cmems_mod_nws_phy-uv_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_CUR_mm{{yyyymm}}.nc | 7.1 |
| cmems_mod_nws_phy-mld_my_7km-2D_P1M-m | metoffice_foam1_amm7_NWS_MLD_mm{{yyyymm}}.nc | 0.2 |
| cmems_mod_nws_phy-s_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_SAL_mm{{yyyymm}}.nc | 2.6 |
| cmems_mod_nws_phy-ssh_my_7km-2D_P1M-m | metoffice_foam1_amm7_NWS_SSH_mm{{yyyymm}}.nc | 0.2 |
| cmems_mod_nws_phy-t_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_TEM_mm{{yyyymm}}.nc | 3.2 |
| cmems_mod_nws_phy-bottomt_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_BED_hi{{yyyymmdd}}.nc | 4.4 |
| cmems_mod_nws_phy-mld_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_MLD_hi{{yyyymmdd}}.nc | 4.8 |
| cmems_mod_nws_phy-sss_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_SSS_hi{{yyyymmdd}}.nc | 3.3 |
| cmems_mod_nws_phy-ssh_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_SSH_hi{{yyyymmdd}}.nc | 4.4 |
| cmems_mod_nws_phy-sst_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_SST_hi{{yyyymmdd}}.nc | 4 |
| cmems_mod_nws_phy-uv_my_7km-2D_PT1H-i | metoffice_foam1_amm7_NWS_SSC_hi{{yyyymmdd}}.nc | 10 |
| cmems_mod_nws_phy_my_7km-3D_static | NWS-MFC_004_009_static.nc | 11 |

| NWSHELF_MULTIYEAR_BGC_004_011 | | |
|---|---|---|
| **DATASET NAME** | **NAME OF FILE** | **DIMENSION [MB]** |
| cmems_mod_nws_bgc-kd_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_ATTN_dm{{yyyymmdd}}.nc | 0.7 |
| cmems_mod_nws_bgc-chl_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_CPWC_dm{{yyyymmdd}}.nc | 1.4 |
| cmems_mod_nws_bgc-o2_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_DOXY_dm{{yyyymmdd}}.nc | 2.6 |
| cmems_mod_nws_bgc-no3_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_NITR_dm{{yyyymmdd}}.nc | 1.9 |
| cmems_mod_nws_bgc-po4_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_PHOS_dm{{yyyymmdd}}.nc | 1.7 |
| cmems_mod_nws_bgc-phyc_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_PHYT_dm{{yyyymmdd}}.nc | 1.2 |
| cmems_mod_nws_bgc-pp_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_PPRD_dm{{yyyymmdd}}.nc | 1.3 |
| cmems_mod_nws_bgc-spco2_my_7km-2D_P1D-m | metoffice_foam1_amm7_NWS_PCO2_dm{{yyyymmdd}}.nc | 0.1 |
| cmems_mod_nws_bgc-ph_my_7km-3D_P1D-m | metoffice_foam1_amm7_NWS_PHPH_dm{{yyyymmdd}}.nc | 0.5 |
| cmems_mod_nws_bgc-kd_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_ATTN_mm{{yyyymm}}.nc | 2.4 |
| cmems_mod_nws_bgc-chl_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_CPWC_mm{{yyyymm}}.nc | 2.6 |
| cmems_mod_nws_bgc-o2_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_DOXY_mm{{yyyymm}}.nc | 3.2 |
| cmems_mod_nws_bgc-no3_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_NITR_mm{{yyyymm}}.nc | 3.2 |

| | | |
|---|---|---|
| cmems_mod_nws_bgc-po4_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_PHOS_mm{{yyyymm}}.nc | 3.1 |
| cmems_mod_nws_bgc-phyc_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_PHYT_mm{{yyyymm}}.nc | 2.4 |
| cmems_mod_nws_bgc-pp_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_PPRD_mm{{yyyymm}}.nc | 2.4 |
| cmems_mod_nws_bgc-spco2_my_7km-2D_P1M-m | metoffice_foam1_amm7_NWS_PCO2_mm{{yyyymm}}.nc | 0.2 |
| cmems_mod_nws_bgc-ph_my_7km-3D_P1M-m | metoffice_foam1_amm7_NWS_PHPH_mm{{yyyymm}}.nc | 1.3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-diato_P1D-m | metoffice_foam1_amm7_NWS_DIATO_CPWC_dm{{yyyymmdd}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-dino_P1D-m | metoffice_foam1_amm7_NWS_DINO_CPWC_dm{{yyyymmdd}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-nano_P1D-m | metoffice_foam1_amm7_NWS_NANO_CPWC_dm{{yyyymmdd}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-pico_P1D-m | metoffice_foam1_amm7_NWS_PICO_CPWC_dm{{yyyymmdd}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-diato_P1M-m | metoffice_foam1_amm7_NWS_DIATO_CPWC_mm{{yyyymm}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-dino_P1M-m | metoffice_foam1_amm7_NWS_DINO_CPWC_mm{{yyyymm}}.nc | 3 |
| cmems_mod_nws_bgc-pft_my_7km-3D-nano_P1M-m | metoffice_foam1_amm7_NWS_NANO_CPWC_mm{{yyyymm}}.nc | 3 |

| | | |
|---|---|---|
| cmems_mod_nws_bgc-pft_my_7km-3D-pico_P1M-m | metoffice_foam1_amm7_NWS_PICO_CPWC_mm{{yyyymm}}.nc | 3 |
| cmems_mod_nws_bgc_my_7km-3D_static | NWS-MFC_004_011_static.nc | 11 |

### III.4 Remember: scale_factor & add_offset / missing_value / land mask

Real_Value = (Display_Value X scale_factor) + add_offset (only for NWSHELF_MULTIYEAR_PHY_004_011)

The missing value for this product is: -32768s for NWSHELF_MULTIYEAR_PHY_004_011 and -32768.f for NWSHELF_MULTIYEAR_BGC_004_011

Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

### III.5 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

## IV REFERENCES

Quality Information Documents
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-009.pdf
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-011.pdf

Flather R.A., (1976). A tidal model of the north west European continental shelf. Memoires de la Societe Royale de Sciences de Liege,6,141-164.

Flather R.A., (1981). Practical surge prediction using numerical models. In: Peregrine DH (ed) Floods due to high winds and tides. Academic Press, London New York.

Kara A.B.,.Rochford P.A, Hurlburt H.E., (2000). An optimal definition for ocean mixed layer depth, J. Geophys. Res., 105,16803-16821, https://doi.org/10.1029/2000JC900072

Lenhart H.-J., Mills D.K., Baretta-Bekker H., van Leeuwen S.M., van der Molen J., Baretta J.W., Blaas M., Desmit X., Kühn W., Los H.J., Ménesguen A., Neves R., Proctor R., Ruardij P., Skogen M.D., Vanhoutte-Brunier A., Villars M.T., Wakelin S., (2010). Predicting the consequences of nutrient reduction on the eutrophication status of the North Sea, Journal of Marine Systems, Vol 81,148-170.

Vörösmarty J., Green P., Salisbury J., Lammers R. B., (2000). Global Water Resources: Vulnerability from Climate Change and Population Growth, Science, 289, 284–288, https://doi.org/10.1126/science.289.5477.284

Young E.F, Holt J.T., (2007). Prediction and analysis of long-term variability of temperature and salinity in the Irish Sea. Journal of Geophysical Research, 112 (C1). Art. No. C01008. https://doi.org/10.1029/2005JC003386

## V ANNEX

```
netcdf metoffice_foam1_amm7_NWS_SSH_dm20180301 {
dimensions:
        time = UNLIMITED ; // (1 currently)
        latitude = 375 ;
        longitude = 297 ;
variables:
        double time(time) ;
                time:axis = "T" ;
                time:calendar = "Gregorian" ;
                time:long_name = "Validity time" ;
                time:standard_name = "time" ;
                time:units = "seconds since 1970-01-01 00:00:00" ;
        float latitude(latitude) ;
                latitude:standard_name = "latitude" ;
                latitude:long_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:nav_model = "grid_T" ;
        float longitude(longitude) ;
                longitude:standard_name = "longitude" ;
                longitude:long_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:nav_model = "grid_T" ;
        short zos(time, latitude, longitude) ;
                zos:_FillValue = -32768s ;
                zos:missing_value = -32768s ;
                zos:scale_factor = 0.001f ;
                zos:add_offset = 0.f ;
                zos:standard_name = "sea_surface_height_above_geoid" ;
                zos:long_name = "Sea surface height above geoid" ;
                zos:units = "m" ;
```

```
                zos:valid_min = -10000s ;
                zos:valid_max = 10000s ;

// global attributes:
                :title = "daily-mean SSH (2D)" ;
                :Conventions = "CF-1.7" ;
                :references = "http://marine.copernicus.eu/" ;
                :institution = "UK Met Office" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :netcdf-version-id = "netCDF-4" ;
                :creation_date = "2021-02-08T00:00:00Z" ;
                :source = "AMM-FOAM 7 km (tidal) NEMO v3.6_FABM-ERSEM v15.06_NEMOVAR v6" ;
                :product = "NWSHELF_MULTIYEAR_PHY_004_009" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :forcing_data_source = "ECMWF; ERA5" ;
                :licence  =  "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence/" ;
                :history = "See source and creation_date attributes" ;
```